

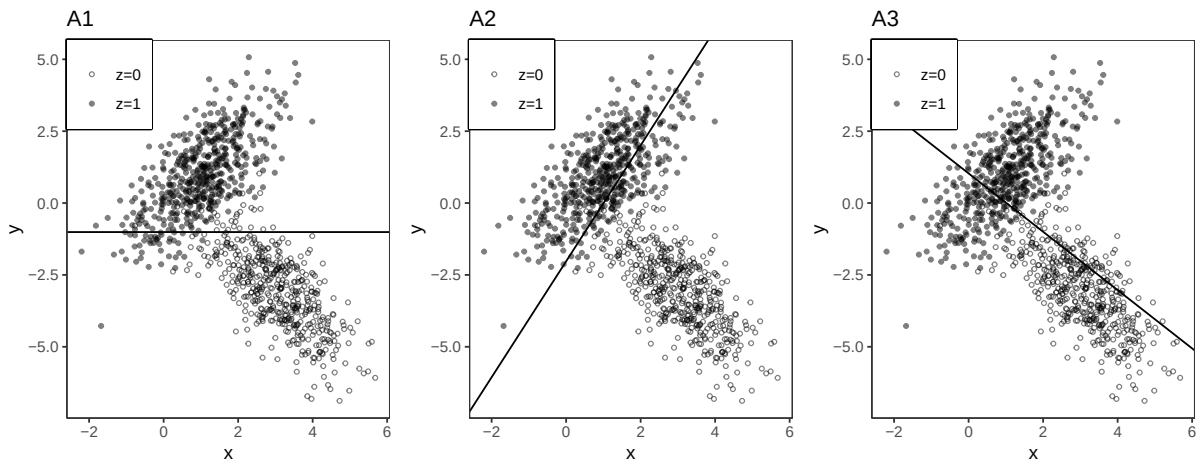
Solutions: Midterm 2 Practice Problems

Problem 1

In the following three parts, choose the figure with the appropriate regression line(s).

Part A

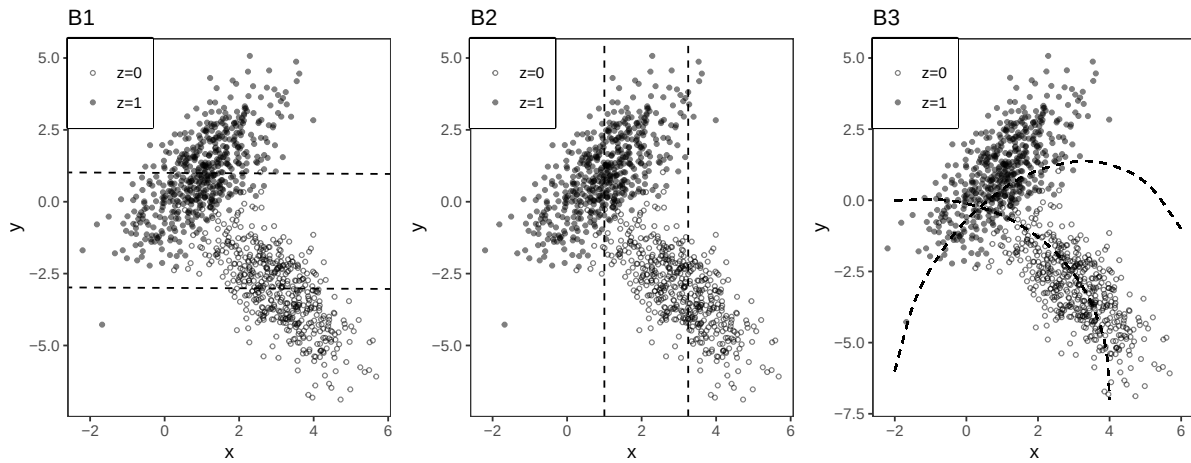
Choose the figure with the solid line that represents the simple regression line ($y \sim x$).



A3

Part B

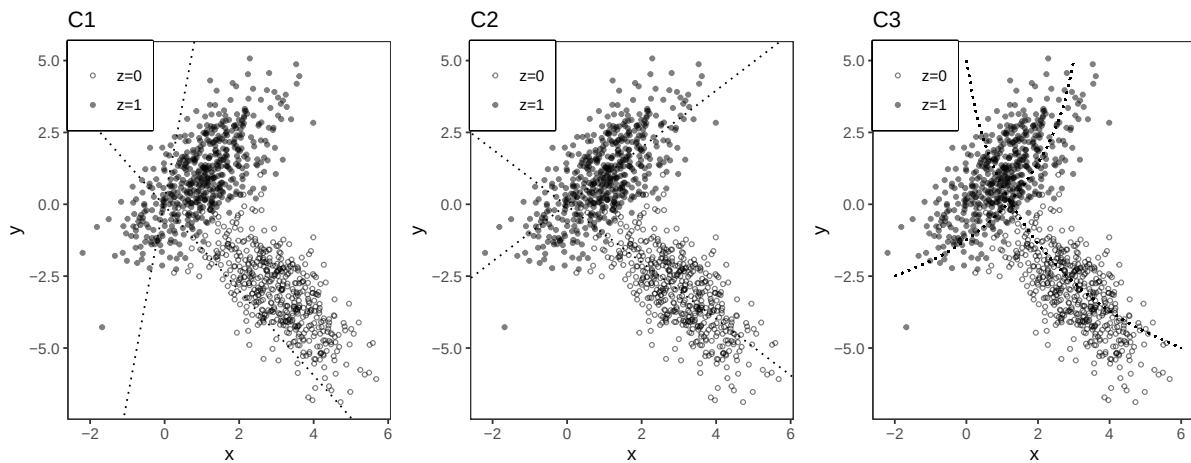
Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



B1

Part C

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



C2

Problem 2

Suppose we have a sample $Y_1 \dots Y_n$ stored on an $\mathbf{R}$ vector Y of length n . We run this code.

```

B=10000
Y.bar = rep(NA,B)
for(b in 1:B) {
  Y.bar[b]= mean(sample(Y,size=n,replace=TRUE))
}

```

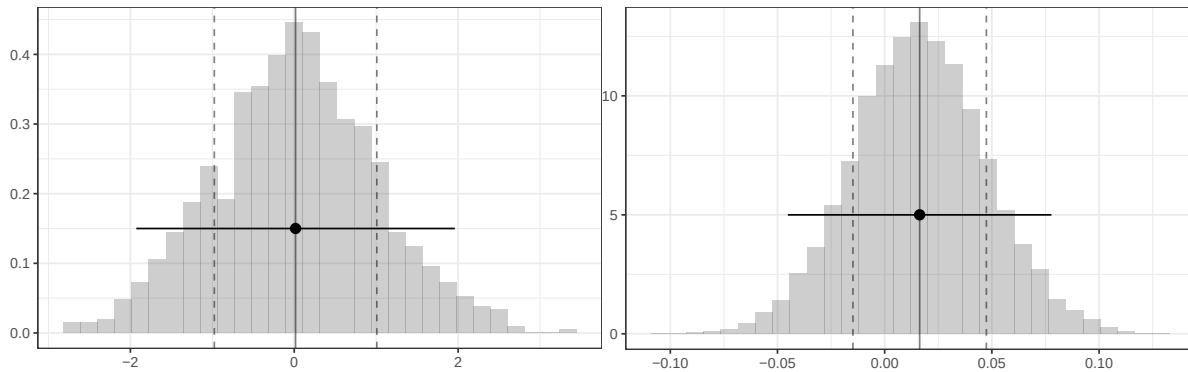


Figure 1: Histogram of the vector Y's elements Figure 2: Histogram of the Y.bar's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the left histogram.

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation. It's the distance between the solid vertical and either dashed vertical line on the left histogram.

Part c

What does this code calculate? State it in words and mark it with an ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y.bar)
```

The bootstrap sampling distribution’s standard deviation. It’s the distance between the solid vertical and either dashed vertical line on the left histogram.

Part d

What does this code calculate? State it in words and mark it with an ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y) - 1.96* sd(Y)/sqrt(n)
mean(Y) + 1.96* sd(Y)/sqrt(n)
```

A 95% confidence interval — based on normal approximation — for the mean of the population our sample $Y_1 \dots Y_n$ is drawn from. This is indicated by the horizontal segment (with a dot in the middle) on the right histogram.

Part e

How could you calculate roughly the same thing as in part d using ‘sd(Y.bar)’ and not ‘sd(Y)’?

We can use the standard deviation of the bootstrap sampling distribution, rather than ‘sd(Y)/sqrt(n)’, as our estimate of standard error.

```
mean(Y) - 1.96 * sd(Y.bar)
mean(Y) + 1.96 * sd(Y.bar)
```

Problem 3

Consider the following R code:

```
1. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(z~x)))
3. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
4. lm(residuals(lm(y~z)) ~ x)
5. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
6. lm(residuals(lm(y~x)) ~ z)
7. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
8. lm(y ~ residuals(lm(x~z)))
```

a) Which one or more of the above will produce the coefficient on x from the multiple regression of $y \sim x + z$?

3 and 8

b) Which one or more of the above will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$?

3 and 7